

CSE 6730 Modeling and Simulation: Foundations and Implementation

Spring 2026

Instructor: Spencer H. Bryngelson

Final Project

Georgia Tech Campus Crowd Simulation

Group # 17

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May 1st, 2026

Link to GitHub Repository

<https://github.gatech.edu/acheng324/GaTechCrowdSimulator>

1. Abstract

Our project aims to study pedestrian dynamics in multiple scenarios on the Georgia Tech campus, via the use of a pedestrian dynamics model. In particular, we make use of the Social Force model (Helbing & Molnar, 1995) and take into account path constraints. We aim to study pedestrian behaviors in scenarios focused on “multi-source, single-sink” (simulating events on campus, e.g. sporting events or concerts). By analyzing these scenarios, we hope to better understand pedestrian dynamics in high-density, bounded environments and how key bottlenecks in traffic and congestion emerge, as well as how thoughtful urban design can mitigate these scenarios.

2. Project description

The goal of our project is to model and simulate pedestrian movement on the Georgia Tech campus, with a focus on high-density event scenarios. We are especially interested in understanding where congestion may occur, how pedestrians move through the campus path network, and which sidewalks, intersections, or entrances may become bottlenecks. Our main event scenario studies pedestrian traffic before and after a large event at Bobby Dodd Stadium, where visitors arrive from parking lots or MARTA stations and walk toward the stadium. This follows our project framing of Georgia Tech as a relatively bounded, walkable, high-density campus environment.

In our model, each pedestrian is represented as an individual moving agent. Each agent has a starting location, destination, walking speed, and route. The campus is represented as a simplified pedestrian path network, so agents move along realistic walkable routes rather than freely across the map. Shortest paths between destinations are computed using Dijkstra’s algorithm to guide agents from origins to destinations.

The most important aspects to capture are pedestrian origins and destinations, campus pathway geometry, walking speed, crowd density, route choice, and interactions between pedestrians. These are especially important during event-based “bursts” of movement, when many people enter or leave the stadium at the same time. Such bursts can create congestion at narrow paths, corners, intersections, and entrances. Our model primarily presents this pre-event arrival simulation as a campus case study.

After conducting a review of the diverse literature written on modeling pedestrian traffic, we decided that the most important models are ones that capture realistic pedestrian behaviors, most notably density-awareness. Due to its intuitive theory and parameters, we base our simulation from the Social Force Model (Helbing & Molnar, 1995). In simple terms, pedestrians are pulled toward their destinations while also being pushed away from nearby people and obstacles. This allows the model to represent goal-directed walking, collision avoidance, and personal space. Our initial test showed that attraction alone can cause agents to overlap, while adding repulsion produces more realistic crowd movement. We extend this to Georgia Tech’s closed-path system in this final submission.

Overall, the project aims to combine campus-scale path modeling with individual pedestrian behavior. Our final simulation helps visualize pedestrian traffic intensity, identify likely bottlenecks, and compare movement patterns across different scenarios.

3. Literature review

Existing Pedestrian Simulation Models

Modeling pedestrian movement poses an interesting and unique scenario for simulation tasks, as it offers the designer plenty of creativity in the underlying physical principles behind pedestrian movement, and can serve

as a powerful tool in predicting how inhabitants move throughout a space. The existing literature poses a variety of methods for simulating such movement based on the level of detail one wishes to model. The Hughes Model consists of a series of interrelated PDEs that govern pedestrian density throughout a continuous 2D domain and time sequence (Hughes, 2000). This model simulates the large-scale flow of pedestrians from one area to another, centered around the conservation law, $\partial_t \rho - \Delta(\rho v) = 0$, which also governs optimal direction and travel time potential based on density.

Hughes Model has been adapted to fit discrete 2D domains, which is suitable to a situation where the 2D domain is representable as a graph where pedestrians can only occupy a fixed path of edges and nodes (Camilli et al., 2016).

Unlike Hughes Model which focuses on macrocosmic, large-scale pedestrian flow, there are also finer-grained agent-based simulations that model pedestrian environments based on the probabilistic behaviors of the individual agents. One such work includes the Social Force Model (Helbing & Molnar, 1995), which governs pedestrian movement towards a destination subject to speed sampled from a probabilistic distribution with an optimal direction (Kretz et al., 2024), stochastic fluctuations in movement, as well as constraints such as attractive forces (friends) and repulsive forces (building edges, personal space, large crowds). Behavior Models offer a way to represent realistic pedestrian choices beyond just shortest-path motivations, expanding the choice of behaviors to include choosing from various transportation modes, departure times, or even the choice to reconsider going to an event at all (Train, 2009, pp. 15-40). By balancing an "attractive" force pulling individuals toward their goal with "repulsive" forces pushing them away from obstacles and peers, Helbing & Molnar's model showed that emergent macroscopic crowd behaviors, like lane formation in hallways, could be simulated using microscopic continuous equations. The extended social force model (Jiang et al., 2017) extends Helbing & Molnar's model to include density-awareness, contributing to the high-congestion avoidance seen in large crowds. Such models succeed at capturing a more realistic distribution of pedestrian movement, but are difficult to scale to large populations.

To simulate larger crowds more efficiently, researchers also explored discrete mathematics, such as the cellular automata microsimulation proposed by Blue and Adler (2001). By dividing walkways into uniform grids and dictating movement through strictly local, probabilistic rules, this approach demonstrated that fundamental traffic properties could be accurately reproduced without complex mathematics, offering a highly computationally efficient alternative. Building on efficient navigation, agent-based kinematics advanced significantly with the Optimal Reciprocal Collision Avoidance (ORCA) framework by van den Berg et al. (2011). ORCA framed multi-agent collision avoidance as a linear programming problem, providing mathematically collision-free navigation for large numbers of dense, independent agents under strict computational constraints.

While geometric and physics-based models provided robust mathematical foundations, they often lacked psychological realism, prompting the integration of cognitive science. Moussaïd, Helbing, and Theraulaz (2011) addressed this by replacing simulated physical "forces" with cognitive, vision-based behavioral rules. Their model established that under normal conditions, pedestrians use a visual heuristic: they look ahead, calculate the "time to collision" for various potential paths, and continuously choose the most direct, obstacle-free route. Crucially, this framework unified everyday walking dynamics with extreme crowd disasters by incorporating a threshold where cognitive rules fail and unintentional physical bodily forces take over during severe congestion.

Most recently, the field has experienced a paradigm shift toward data-driven models. Alahi et al. (2016) pioneered this with Social LSTM, utilizing Long Short-Term Memory neural networks to predict future pedestrian trajectories based on historical movement data. Through a novel "Social Pooling" layer, spatially

proximal networks could share hidden states at each time step, allowing the model to implicitly learn realistic collision avoidance and human interaction rules directly from empirical data rather than relying on manually defined equations.

Pedestrian Dynamics Simulation Scenarios and Applications

There are a wide variety of scales and scenarios that have been used to study pedestrian dynamics. One example is at the micro-scale (Jain et al., 2004). The authors observe that most research in this domain is focused on modeling crowded situations, and not common medium-to-low density situations. Therefore, they evaluate multiple force-based models on “micro-scale” situations such as navigating around a stationary obstacle, and around another agent.

Another example is city-scale modeling of pedestrian dynamics, as seen in Kazyieva et al., 2023. The authors develop a model for regional-scale pedestrian dynamics and perform a case study on the city of Salzburg in Austria. They take into account different kinds of activities, and also different kinds of pedestrians (e.g. residents and tourists). The different kinds of pedestrians are restricted on different dimensions. For example, residents are restricted to a set amount of activities per day, and tourists are restricted either based on time or distance. For realism, the authors use several open resources to obtain data about the city and its residents.

Zooming in from city-scale dynamics, another scenario that has been explored is that of commuters in public transportation hubs. Data on pedestrian dynamics patterns can help planners design to mitigate effects like congestion. Zhen et al., 2024 analyze the 59th Street and Columbus Circle transfer hub in the New York City subway system. They use AnyLogic, a simulation software, in conjunction with data on train arrival times, to construct a model that simulates passengers arriving at the station and switching lines. The authors uncovered multiple regional overcrowding spots that signalled improvements could be made to the station.

Another set of scenarios is based around setting environmental conditions, and evaluating how these affect pedestrian movement. Papers like Melnikov et al., 2022 explore how temperature affects factors like path choices. They use both experimental and computational approaches to build a model that takes into account both shading and distance considerations. From their observations, they are able to uncover a perceived cost of walking under the sun, and take into account the fact that tree shade is different from building shade. Such environmental-based models could prove to be valuable in urban pedestrian simulations, as they add another layer of nuance to pedestrian behavior.

Pedestrian Dynamics Simulation Platforms and Tools

Understanding urban pedestrian dynamics relies heavily on pedestrian simulation software. Such software generally operates in three phases: data preprocessing, simulation engine execution, and output visualization. When selecting a simulation platform, one should balance the computational demands, accuracy of behavior, and granularity needed for research objectives.

As shown in Table 1, there are multiple choices for simulators, each with their own unique profiles. Some, like SUMO and VISSIM, are mainly traffic-oriented and focus on vehicles, signals, and infrastructure control. In contrast, platforms such as LEGION and Vadere concentrate on pedestrian behavior and crowd dynamics. They are often used for studies on evacuation, navigation, and high-density urban flows. For large pedestrian populations, scalable simulators like FLAME GPU and Repast HPC employ force-based or distributed agent models to accurately represent movement at scale (Zambare et al., 2025).

Table 1 Comparison of pedestrian simulation and capability overview

Software	Primary Simulation	Route Choice	Use-Case Strength	Extensibility	Pedestrian Model	License
<i>SUMO</i>	Traffic/Pedestrian	Dynamic shortest path, rerouting, lane selection	Multimodal, autonomous driving, large urban traffic	High (250+ tools, APIs, TraCI)	Agent-based, rule-based, extensible with force models	Open
<i>VISSIM</i>	Integrated Traffic/Pedestrian	Rule-based, LOS decisions, probabilistic paths	Urban operations, policy testing, transit/shared spaces	Moderate (Proprietary COM API)	Calibrated Social Force Model	Commercial
<i>Vadere</i>	Pedestrian/Crowd	Model-dependent (heuristics, gradient)	Locomotion comparison, evacuation, interdisciplinary	High (Modular Java, open-source plugins)	Optimal Steps, Gradient Navigation, Social Force Models	Open
<i>JuPedSim</i>	Pedestrian/Crowd	Multi-stage, conditional routing via deciders	Evacuation planning, building safety, experimental replication	High (Python/C++ core, customizable agents)	Collision-Free Speed, Generalized Centrifugal Force Models	Open
<i>FLAME GPU</i>	Crowd/Pedestrian (at scale)	Force-based, goal-directed	Large-scale crowd dynamics, GPU behavioral studies	High (C++/CUDA, Python APIs, customizable logic)	Built-in social force-based	Open
<i>Repast HPC</i>	Large-scale Agent-based	User-defined (custom path planning via agent logic)	Massive crowd modeling, urban mobility	High (C++, MPI, customizable behaviors)	Customizable (supports force-based via user logic)	Open

4. Conceptual model

We chose to implement a simple Social Force model (Helbing & Molnar, 1995). In this paper, pedestrians are modeled as individual forces subject to several force terms represented as a system of ODEs in which you can solve for either an attractive force term or a repulsive force term. We chose this microscopic model as a starting point as it attempts to simulate the system as interactive individual behaviors and is thus more intuitive to understand. Compared to modeling the environment as literally external forces that act on an agent's body, the environment affects their "motivation," which manifests itself in changes in acceleration.

Given a 2D space with unconstrained paths, a closed system of randomly positioned agents, and one destination, we can model the total "social force" acting on any particular pedestrian's (α) movement as the sum of attractive forces (from the goal) F_{α}^0 , and repulsive forces (against the personal space of other agents)

$f_{\alpha\beta}$:

$$F_{\alpha}(t) = F_{\alpha}^0(v_{\alpha}, v_{\alpha}^0 e_a) + \sum_{\beta} f_{\alpha\beta}(r_{ab})$$

The force from the destination is modeled as $F_{\alpha}^0(v_{\alpha}, v_{\alpha}^0 e_a) = \frac{1}{\tau} (v_{\alpha}^0 e_a - v_{\alpha})$, where $v_{\alpha}^0 e_a$ represents a velocity vector of desired, unconstrained speed in the desired direction (towards the goal), v_{α} is the current velocity, and τ represents the time it takes to obtain that speed.

The repulsive force from another agent, β , acting on α , is represented as $f_{\alpha\beta}(r_{ab}) = -\Delta_{r_{\alpha\beta}} V_{\alpha\beta}[b(r_{\alpha\beta})]$, where V is a monotonically decreasing function as a function of a private sphere ellipse defined by b , which defines how much agent β repels α based on β 's velocity and position (e.g. humans use others' velocities to avoid collisions, and position since we only care about distance from people in front, not behind).

As the Social Force paper assumes an unconstrained 2D environment, the desired direction cannot be a simple vector from agent to destination. We use Dijkstra's shortest path algorithm to compute the ideal route between each agent and the destination, then iteratively recompute desired direction based on where the path allows us to go. Additionally, we use logic from the Extended social force paper (Jiang et al., 2017) to include density awareness, such that agents will allow themselves to take longer paths if it means avoiding a high congestion area. We model density as, $\rho^k(r) = \sum_i f(|r_i^k - r|)$, where we add a penalty term for the cost of a particular path for an agent based on the number of agents in that agent's subarea.

The above being the key modeling forces in social force model, we opted not to use several of the other forces mentioned, such as any attractive forces between pedestrians (representing friendship interactions), stochastic differences in agent speed and behavior (like taking detours), and repulsion from the wall (though that logic is commented out in the codebase).

We've also decided to not use pre-existing simulation software as there are far more variables than we would be able to account for in the given scope of this project, and the ability to experiment with their existing code is more difficult than creating a simulation from scratch.

5. Simulation - simulator / simulation model

We implement the social force mathematics described in the section above and apply it to our simplified campus map. The model has multiple parameters that can be toggled on or off depending on what functionality the user would like to emphasize, which made the social force model intuitive and easy to understand, as the parameters reflect real-world agent behaviors, rather than high-level physics concepts. While we implemented the above equations, we had to introduce computational parameters as computing density fields for every agent became an unwieldy operation to do every time step. Instead, we introduce discretization variables that allow us to calculate density at coarser grids and replan in longer interval time steps, which helped speed up the simulation. We then apply this logic to our campus scenario in the next section.

Bobby Dodd Stadium Pre-Event Arrival Simulation Parameters and Configuration

This simulation models pedestrian movement during large-scale events at Bobby Dodd Stadium at Hyundai Field, which has a capacity of over 50,000 people and hosts events such as football games and concerts. The main focus is the pre-event arrival phase, where people walk from the surrounding parking lots to the stadium entrances.

- (1) Simulation Environment Setup. As shown in Figure 1, the simulation environment includes the stadium and its surrounding infrastructure, specifically 18 parking lots. The stadium has 4 entrance gates, while the parking lots together include 37 access gates. The model assumes a total of 30,000 attendees arrive by car, park in these lots, and then walk toward the stadium. In addition, Gate 21 is treated as both a parking gate and a metro access point, since it is located close to the North Avenue subway station. This allows the model to capture both driving and public transit arrivals.

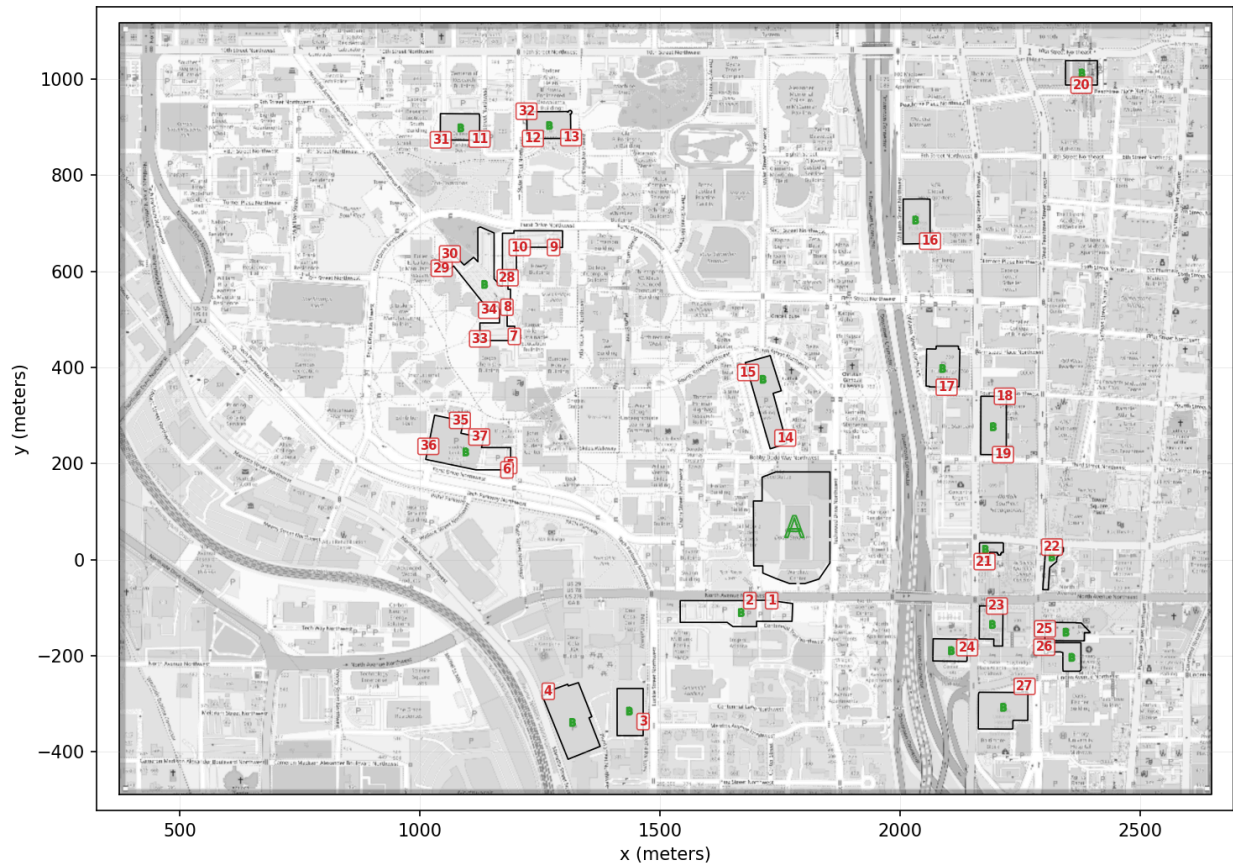


Figure 1. Simulation environment

- (2) Agent Initialization. We assume most people will prefer three main options: taking the metro, parking close to the stadium, or parking at large parking lots. Based on this assumption, parking lots are grouped into four categories, and each category is assigned a different number of agents to reflect a more realistic agent distribution. Table 2 summarizes the initial number of agents assigned to each gate.

Table 2. Agent initialization by gate categories

Group	Gate Numbers	People per Gate	Total
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Metro Station	21	7,500	7,500
Near Stadium	1, 2, 14, 15	1,000	4,000
Large Parking Area 1	5, 6, 35, 36, 37	1,000	5,000
Large Parking Area 2	7, 8, 9, 10, 28, 29, 30, 33, 34	700	6,300
Distant Parking Areas	Remaining 18 gates	400	7,200

- (3) Constrained Scenarios. In addition to the above, base scenario, we study constrained scenarios that model how the system would behave if there were a restriction on available parking lots. We keep the total number of agents to around 30,000 for consistency. In our first constrained scenario, the two possible initialization points for agents are a large parking lot and the metro station. In our second constrained scenario, we model if only the metro station and only distant and smaller parking lots are available.

Table 3. Agent initialization in constrained scenario 1

Group	Gate Numbers	People per Gate	Total
Metro Station	21	11,420	11,420
Large Parking Area 1	5, 6, 35, 36, 37	3,920	19,600

Table 4. Agent initialization in constrained scenario 2

Group	Gate Numbers	People per Gate	Total
Metro Station	21	8,305	8305
Distant Parking Areas	18 Gate Numbers excl. all others	1,205	21690

6. Experimental results and validation

Early experimentations in toy environments show that the model was acting in expected ways. Here we can see that introducing repulsive forces made agents respect the personal space of other agents in the vicinity

(table 1). This prevented agents from clustering and touching, and made arrival slower, which we would expect from a real-world scenario.

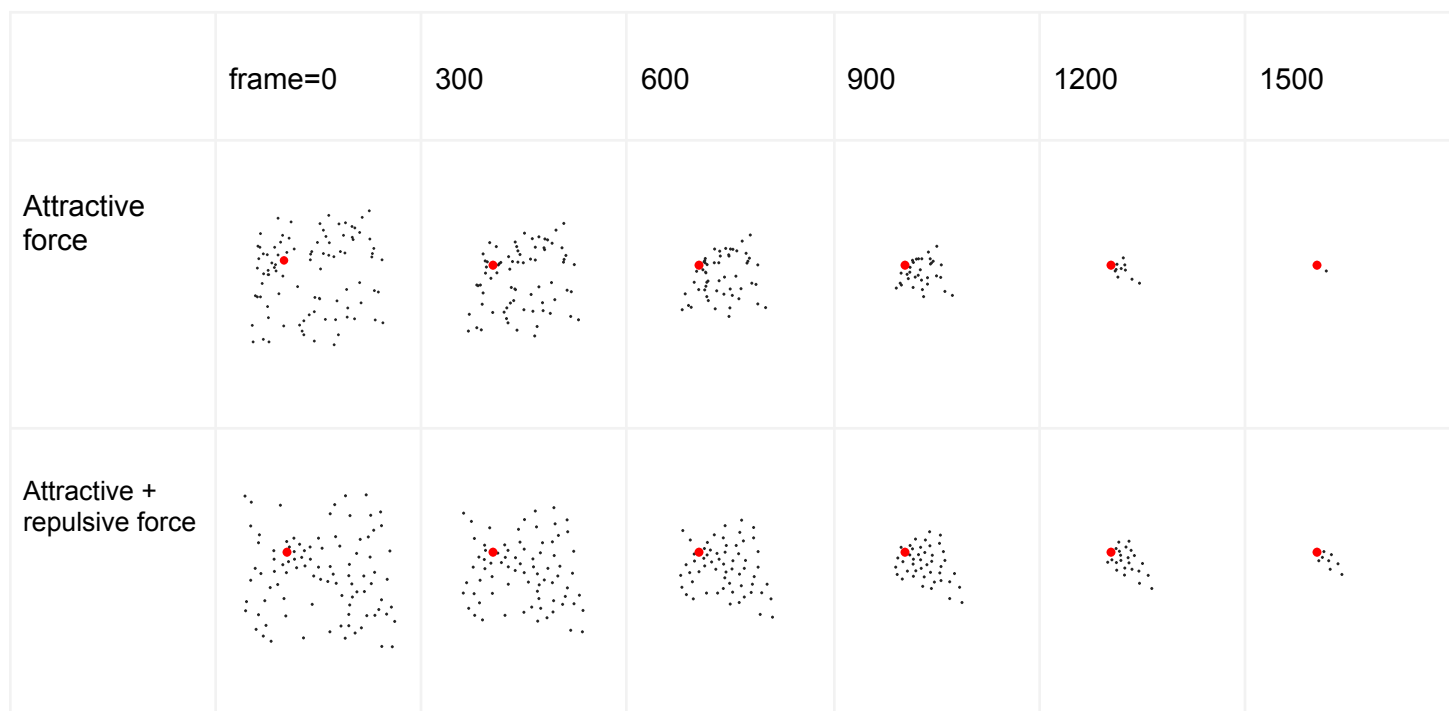


Table 1. Attractive + Repulsive Forces in Unconstrained 2D Domain

Originally, we had also implemented the repulsive effects given by building edges, but opted to drop that term because it produced unrealistic effects when we introduced fixed paths for the agents to walk on. We also noticed that in early simulations, agents would get stuck in queues as all agents in the same path would have the exact same shortest-path calculation, but after introducing density awareness (Jiang et al., 2017), we observed more realistic behaviors of agents avoiding large crowds. We apply this to our final campus map in the next section.

Bobby Dodd Stadium Pre-Event Arrival Simulation Results

As shown in Figure 2, the simulation represents the full pre-event arrival process. We choose to use a logistic curve to model the arrival of agents at different initialization points. Our assumption is that initially a small number of agents will be initialized, modeling early arrivals. We then expect a sharp increase in arrivals (closer to the event start time), and then a leveling off as time goes on. Following this curve, pedestrian flow reaches its peak at around step 1000, which is expected since the logistic model is centered at this point, meaning approximately 50% of agents have been generated. The peak is sustained for about 200 steps, until around step 1200, after which pedestrian flow begins to decline.

Figure 3 shows the highest-density areas during peak flow. The main congestion occurs along Bobby Dodd Way NW and North Ave NW. This pattern is driven by the surrounding access points. Bobby Dodd Way NW receives a large number of pedestrians from the nearby parking lots at Gates 14 and 15. In addition, North Ave NW handles the large number of pedestrians from the metro station. These two roads are also mandatory routes for reaching the stadium entrance gates.

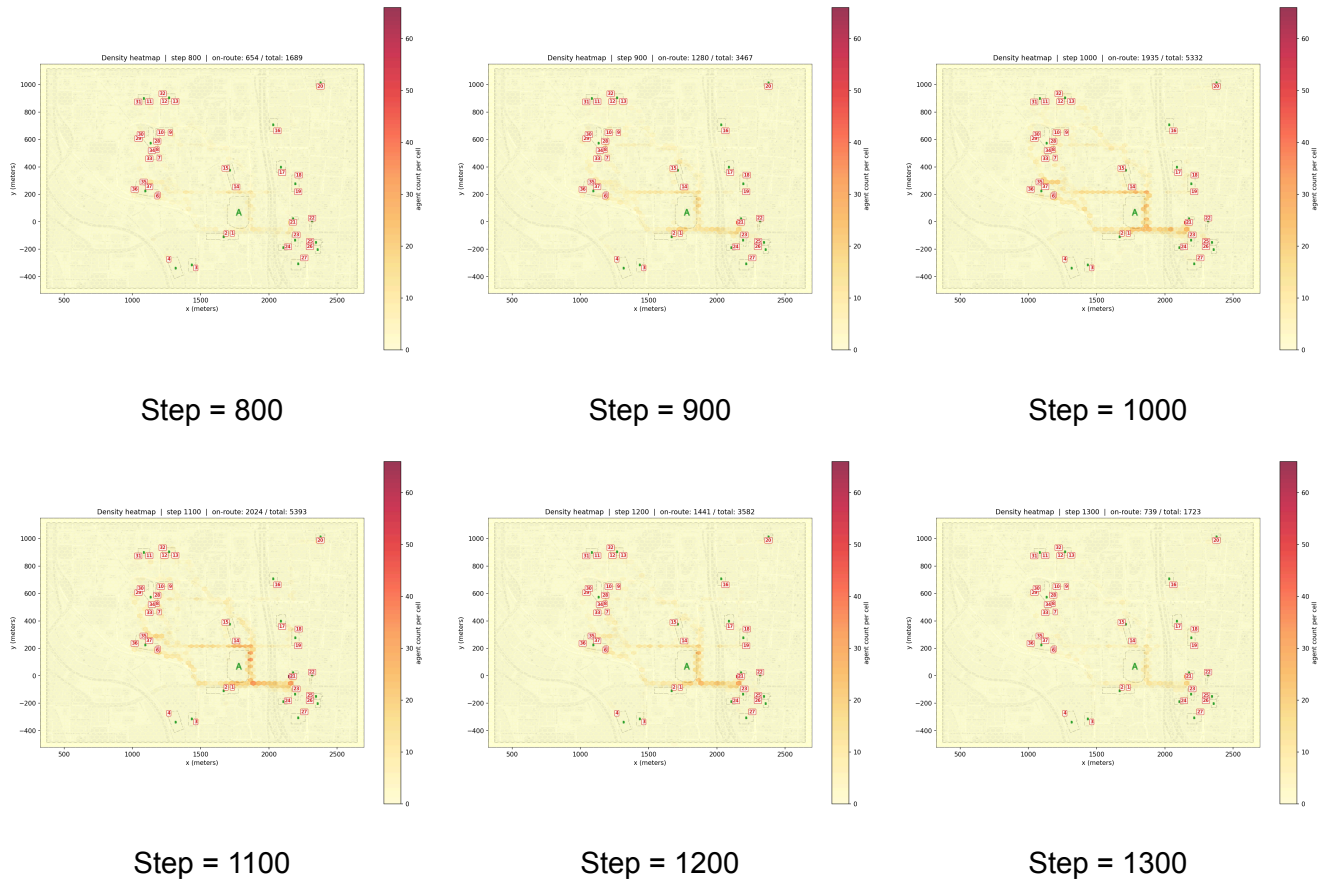


Figure 2. Heatmaps of the base simulation process

Constrained Scenario Results

As seen in Figure 4, in the first constrained scenario, where only a large parking lot and the metro station are available, we can observe similar bottlenecks on North Ave NW and Bobby Dodd Way NW. It is expected that there is congestion on Techwood Dr NW, as this is where the access points to the stadium are. Overall this congestion behavior is expected to be in line with our base simulation, because we simply removed more minor parking lots and reallocated their agents to the available gates. However, in the second constrained scenario, when we restrict available gates to distant parking lots and the metro station, we see some different congestion points. As seen in Figure 5, the three main arteries which pedestrians flow through are Bobby Dodd Way NW (restricted to the eastern side of the stadium, as opposed to the west in the base scenario), North Ave NW, and what can be described as a path through the center of campus. This scenario provides us with an alternative perspective, explicitly modeling how traffic would flow if there were no initialization points near the stadium.

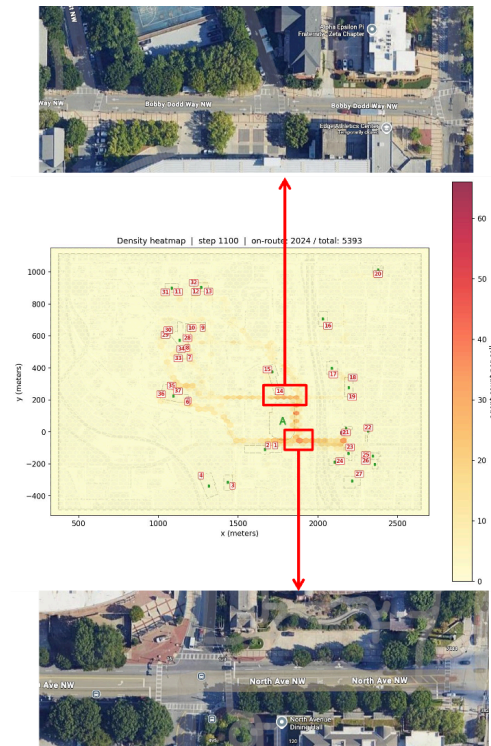


Figure 3. Crowded areas at step 1100

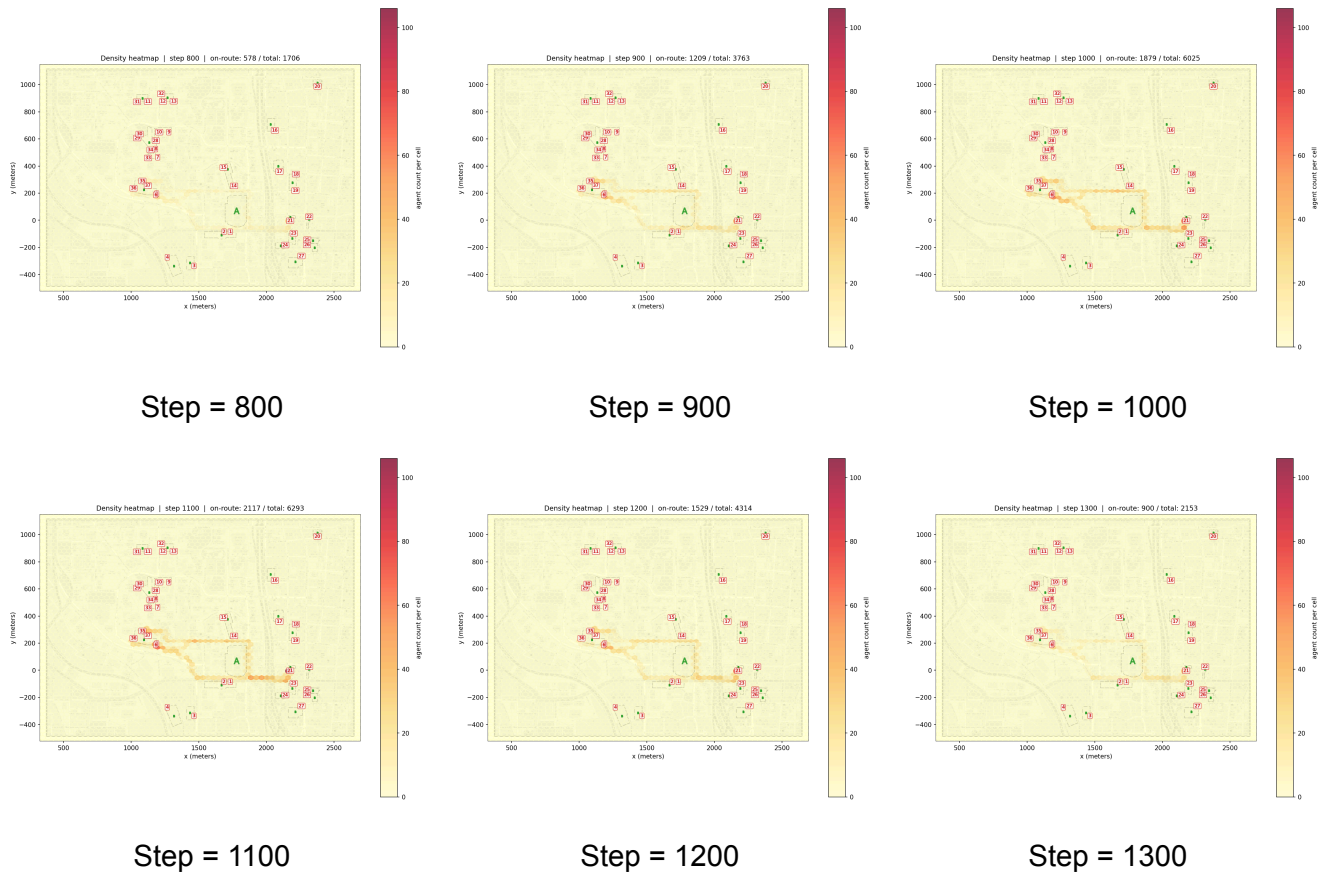


Figure 4. Heatmaps of the constrained scenario 1 (only one large parking lot and metro station)

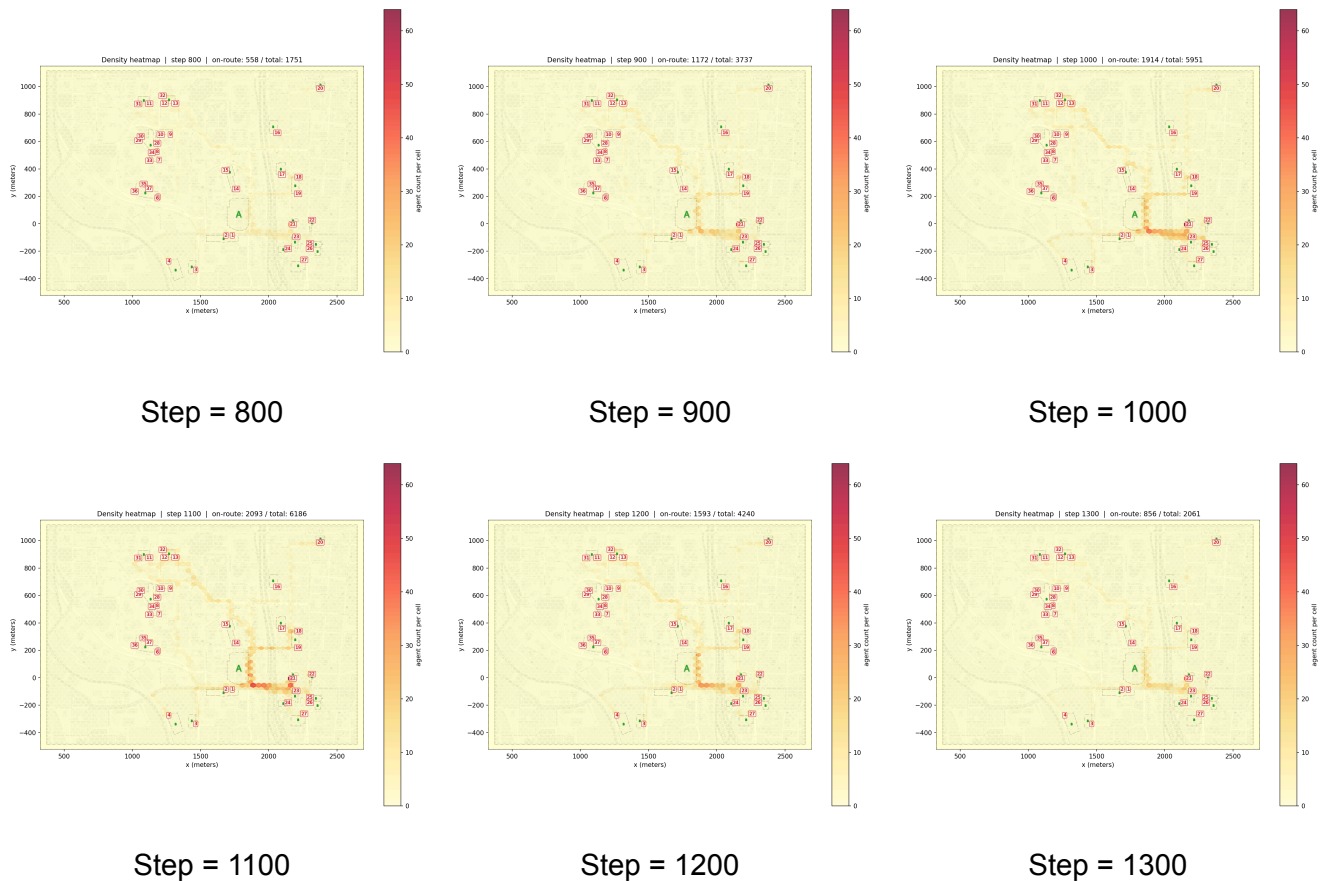


Figure 5. Heatmaps of the constrained scenario 2 (only distant, small lots and metro station)

7. Discussion, conclusions, summary

Through the course of this project, we learned a lot about the design of pedestrian simulations. The process of constructing the simulation map helped us in understanding how to create grounded environments for our experiments. We used open-source map data via OpenStreetMap for the paths and the relevant parking area boundaries, and classified paths into varying levels of bandwidth. One core aspect we picked up on was the fact that we could always improve our simulation to be more realistic, as reflected by our addition of the Social Force model and density-awareness. From our actual simulations, we were able to draw conclusions about congestion points and the paths that would see the most traffic, depending on the locations and initial counts of different initialization areas. In situations with large convergences of people at the Bobby Dodd Stadium, streets such as North Ave NW and Bobby Dodd Way NW will see massive amounts of traffic, as they lie at convergence points of multiple paths from parking areas to the stadium. It would be recommended to place crowd control measures at such areas to mitigate overcrowding.

In terms of next steps, we would focus on modeling “single-source multi-sink” and “multi-source multi-sink” scenarios. The former could correspond to post-event departure behaviors, and the latter could correspond to flows on regular school days. We believe that we could implement a “single-source multi-sink” scenario with some small modifications to our code, since we already have a method of assigning agents to gates. We hypothesize that we would see much of the same behaviors as displayed in our “multi-source single sink” simulations discussed above.

Implementing the “multi-source multi-sink” scenarios would be more challenging, as we would have to generate traces for every single agent. One idea that we had was to scrape enrollment data for various

Georgia Tech courses, and then construct “schedules” for different agents. However, we were not sure how to construct such schedules in a realistic way, on the level of an individual agent. One way would be to construct “profiles” for each agent, following major enrollment distributions at Georgia Tech. Each agent could have assigned attributes such as year of study and program of study, and we could assign schedules to agents based on this information. We would also have to account for behaviors like going to lunch, at dining halls or at restaurants in Tech Square. This would require a great amount of effort that we found to be infeasible given time constraints. We expect that study of such scenarios would reveal even richer patterns that could inform future development of paths and the implementation of congestion accommodating features.

A further level of sophistication would be to include behaviors beyond simply going from point to point. There are often various booths or events on campus that influence the behaviors of students. Another factor could be the presence of friends, e.g. if two students meet on a path, they could stand in the path and talk. Another level of sophistication would be to include varying forms of transportation, e.g. electric scooters or bicycles. It would be interesting to investigate having a non-closed system, where agents could enter or exit the bounds of the simulation at will. One likely implementation instrument for complex planning behaviors would be large language models. Models would be provided with various pieces of context, e.g. energy, stress, assignments, schedules, and relevant buildings within a certain proximity, and they could be free to choose the places to navigate to next. To validate this in a robust manner, we would most likely need to use human subjects as a control, providing them with the same pieces of context, and asking them to take actions. We could then compare LLM-generated actions and human-generated actions to validate if we have achieved an acceptable degree of realism with our LLM-based agents.

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9. Appendix: Division of labor

Alec Cheng	Physical system modeling, codebase manager, simulation implementation.
Jimmy Miao Zhang	Data manager, campus map/plan designer, simulation implementation
Pranav Virupaksha	Physical system modeling, codebase manager
Shiwei Luo	Map and codebase integration, codebase manager